

ANWR TX Cranes	MAE	RMSE	BPR-50	Cook County IL Overdose	MAE	RMSE	BPR-100
Last timestep	0.24	1.04	0.27	Last timestep	1.07	1.66	0.76
Avg over 10	0.25	0.78	0.38	Avg over 5	0.99	1.58	0.80
Chance decision, $\hat{y} = 0$	0.15	0.79	0.05	Chance decision, $\hat{y} = 0$	1.37	2.46	0.20
EpiGNN	0.38	0.70	0.06	EpiGNN	1.33	2.03	0.32
NLL Only	0.35	1.14	0.38	NLL Only	1.03	1.58	0.78
DAML ($\epsilon = 1$)	9.08	35.89	0.39	DAML ($\epsilon = 1$)	1.12	1.84	0.80
BPR Only	40495	40495	0.41	BPR Only	70.31	152.72	0.82

(a) ANWR TX Cranes 2009-2010.

(b) Cook County IL 2021-2022.

Table 1: Performance comparison between different model families for error-based and decision metrics. The top 2 rows represent simple historical baselines: using either the previous timestep alone, or an average over a larger amount (10 bi-months for the crane dataset, and 5 years for Cook County). Next, the *Chance decision* model chooses spatial locations by random chance, and uses all 0's for predicted $\hat{y}$ in MAE and RMSE calculations. Despite the simplicity, this is a competitive model on the error-based metrics due to sparsity. It outperforms all others on MAE on the Cranes dataset. Next is EpiGNN, a spatiotemporal forecasting model based on graph neural networks, trained to minimize RMSE. Although this model has the best performing RMSE on the crane dataset, it makes poor top-K decisions on both. Finally, the last three rows show different ways to train the negative binomial mixed effects regression model in Sec. 5.2. *NLL Only* trains to maximize likelihood alone (Sec. 4.1), *BPR Only* is our direct-loss minimization trained to maximize BPR alone (Sec. 4.2), and DAML (Sec. 4.3) is our hybrid loss that allows a user to explore the pareto frontier between likelihood and BPR-K.